

ECE 3340

Numerical Methods

Exam 2A

Name: _____

Random Numbers _____ / 30

Partial Pivoting _____ / 35

Matrix Conditioning _____ / 35

Total _____ / 100

Random Numbers. Find the maximum period of a 4-bit MLCG with a modulus of 15 and a multiplier of 2.
[15 points]

x_0 must be co-prime with m :

$$\begin{array}{ll} x_0 = 4 & 4 \times 2 \bmod 15 = 8 \\ x_1 = 8 & 8 \times 2 \bmod 15 = 1 \\ x_2 = 1 & 1 \times 2 \bmod 15 = 2 \\ x_3 = 2 & 2 \times 2 \bmod 15 = 4 \\ x_4 = 4 & \end{array}$$

Random Numbers. Calculate the parameters for an LCG that will provide the maximum period for a 3-bit random number generator. Show each value in the sequence starting with $x_0 = 1$. [15 points]

$$m = 8$$

$$c = 1, 3, 5, 7$$

$$a = 5$$

Partial Pivoting and Determinants. Calculate the determinant $|A|$ using partial pivoting. Fill in the blank values at each step. If a vector or matrix doesn't change from one step to the next, you don't have to fill it in (just mark it as the same). **[30 points]**

$$|A| = \begin{vmatrix} 0 & 1 & 5 & 7 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 5 \end{vmatrix} =$$

$$\begin{bmatrix} 0 & 1 & 5 & 7 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 5 \end{bmatrix}$$

$$\ell = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 5 & 7 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 5 \end{bmatrix}$$

$$\ell = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 3 & 6 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

$$\ell = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 3 & 6 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\ell = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

$$\ell = \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

What is $|A^{-1}|$? What formula do you use to calculate this? **[5 points]**

LU Decomposition and Inversion. Calculate the missing column of the matrix inverse given the LU decomposition of a matrix A :

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix}$$

$$U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$$

and **show your work.** [20 points]:

$$\begin{bmatrix} 11/6 & -7/24 & -1/12 \\ -2/3 & 1/12 & 1/6 \\ 1/2 & 1/8 & -1/4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 10 & 6 \\ 3 & 9 & 1 \end{bmatrix}$$

You can check the answer with your calculator. Estimate the condition number for A using the $\| \cdot \|_1$ and $\| \cdot \|_\infty$ norms [15 points].

$$\|A\|_\infty = \max(1 + 2 + 1, 2 + 10 + 6, 3 + 9 + 1) = \max(4, 18, 13) = 18$$

$$\|A\|_1 = \max(1 + 2 + 3, 2 + 10 + 6, 3 + 9 + 1) = \max(6, 21, 8) = 21$$

$$\|A\| = \sqrt{18 \cdot 21} = 19.4$$

$$\|A^{-1}\|_\infty = \max\left(\frac{44}{24} + \frac{7}{24} + \frac{2}{24}, \frac{8}{12} + \frac{1}{12} + \frac{2}{12}, \frac{4}{8} + \frac{1}{8} + \frac{2}{8}\right)$$

$$= \max\left(\frac{53}{44}, \frac{11}{12}, \frac{7}{8}\right) = \frac{53}{44} = 1.20$$

$$\|A^{-1}\|_1 = \max\left(\frac{11}{6} + \frac{4}{6} + \frac{3}{6}, \frac{7}{24} + \frac{2}{24} + \frac{3}{24}, \frac{1}{12} + \frac{2}{12} + \frac{3}{12}\right) = 3$$

$$\|A^{-1}\| = \sqrt{1.2 \cdot 3} = 1.90$$

$$\kappa(A) = 19.4 \cdot 1.20 = 36.86$$